

CSE450 Capstone Project Report

Title of the Project

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CANDIDATES' DECLARATION

We declare that the work presented in this report, titled "Title of the Project", is the outcome of the engineering work carried out by us under the supervision named on the title page.

All prior work, third-party code, libraries, data, copyrighted material, and tools used in the project have been properly acknowledged. Any use of generative AI tools has been disclosed. We accept accountability and responsibility for the accuracy and integrity of this report and for our individual contributions to the project.

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ACKNOWLEDGEMENT

Thank your supervisor, and anyone else who helped: domain experts, collaborators, users who tested the system, people from other departments or organisations you consulted.

ABSTRACT

In no more than one page, state the problem, what was designed and built, how it was verified or validated, and the key outcomes. For a research-oriented project, briefly include the method and principal findings.

Keywords: *three to six keywords*

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Chapter 1

Problem Identification and Formulation

1.1 Background and Motivation

Introduce the domain and explain why the problem matters: who is affected, at what scale, and what the current situation costs them.

1.2 Problem Statement

State precisely the problem the project solves. Formulate it in engineering terms: the inputs and outputs, the entities involved, the quantities to be optimised or guaranteed, and the assumptions made. Identify the engineering knowledge (mathematics, computing fundamentals, specialist knowledge) the problem demands.

1.3 Complexity of the Problem

A complex engineering problem must have characteristic P1 and some or all of P2–P7 (Table 1.1). P1 requires in-depth knowledge at the level of one or more of the knowledge profiles K3–K8 listed in Table 1.2.

For P1, name the knowledge profile(s) involved and show where in the project that knowledge was applied. For each of P2–P7 that applies, give concrete evidence from this project. Drop the rows that do not apply.

1.4 Project Objectives and Scope

State the objectives as a numbered list of concrete, verifiable goals (Chapter 5 evaluates against them), then the scope: what is included and what is explicitly excluded.

Table 1.1: Complex engineering problem characteristics (P1–P7) and evidence from this project

Code	Definition	Evidence from this project
P1	Depth of knowledge required. Cannot be resolved without in-depth engineering knowledge at the level of one or more of K3, K4, K5, K6, or K8, which allows a fundamentals-based, first-principles analytical approach.	<i>Which of K3–K8, and where applied (mandatory)</i>
P2	Range of conflicting requirements. Involves wide-ranging or conflicting technical, engineering, and other issues.	<i>The conflicting requirements and trade-offs</i>
P3	Depth of analysis required. Has no obvious solution and requires abstract thinking and originality in analysis to formulate suitable models.	<i>Why routine methods were insufficient</i>
P4	Familiarity of issues. Involves infrequently encountered issues.	<i>The non-routine issues</i>
P5	Extent of applicable codes. Is outside problems encompassed by standards and codes of practice for professional engineering.	<i>Where standards did not cover the problem</i>
P6	Extent of stakeholder involvement and conflicting requirements. Involves diverse groups of stakeholders with widely varying needs.	<i>The stakeholder groups and their conflicting needs</i>
P7	Interdependence. Is a high-level problem including many component parts or sub-problems.	<i>The sub-problems and their dependencies</i>

Table 1.2: Knowledge profiles referenced by P1

Code	Description
K3	A systematic, theory-based formulation of engineering fundamentals required in the engineering discipline.
K4	Engineering specialist knowledge that provides theoretical frameworks and bodies of knowledge for the accepted practice areas in the engineering discipline; much is at the forefront of the discipline.
K5	Knowledge that supports engineering design in a practice area.
K6	Knowledge of engineering practice (technology) in the practice areas in the engineering discipline.
K8	Engagement with selected knowledge in the research literature of the discipline.

Chapter 2

Preliminary Research

2.1 Literature Review

Review the research literature relevant to the problem: the established methods, their results, and their limitations. Synthesise the findings rather than listing papers one by one, and state what the literature establishes and what it leaves open.

2.2 Existing Solutions and Technologies

Survey the products, systems, and technologies already available in this space, including anything the industry partner already uses: what they do, their strengths, and their limitations. Conclude with the gap this project addresses. Candidate solutions for this specific problem are formulated later, in Section 3.3.

2.3 Market and User Research

Identify who would use the solution and establish the demand for it: the size and characteristics of the user base or market, what users currently do instead, and what they need. Report the evidence gathered (surveys, interviews, discussions with the industry partner, published data). The financial implications are analysed in Section 6.1.3.

2.4 Feasibility Study

Assess whether the project is feasible along the dimensions that matter to it: technical (available methods, data, and computing resources), economic (cost against expected benefit), operational (whether the intended users and organisation can adopt it), and legal or regulatory. State the conclusion and any assumption the project depends on.

Chapter 3

Requirements Analysis

3.1 Project Objectives and Constraints

Identify all stakeholders (users, client or industry partner, administrators, regulators) and their expectations, including conflicting ones. Then clearly state the objectives, the functional and non-functional requirements, and the constraints (time, budget, platform, data, deployment), taking those expectations into account.

3.2 Regulatory Requirements and Standards

Identify all relevant regulatory requirements, engineering standards, and codes of practice (e.g. data-protection law, WCAG, OWASP, ISO/IEC/IEEE standards, industry rules, partner policies). For each, state why it applies and how the design complies.

3.3 Formulation of Solutions

Formulate multiple genuinely different solutions that could each functionally meet most requirements. Compare and evaluate them against explicit criteria from the requirements and constraints, and draw a valid conclusion that justifies the selected solution.

Chapter 4

System Design and Implementation

4.1 Functional Design and Sub-problem Partitioning

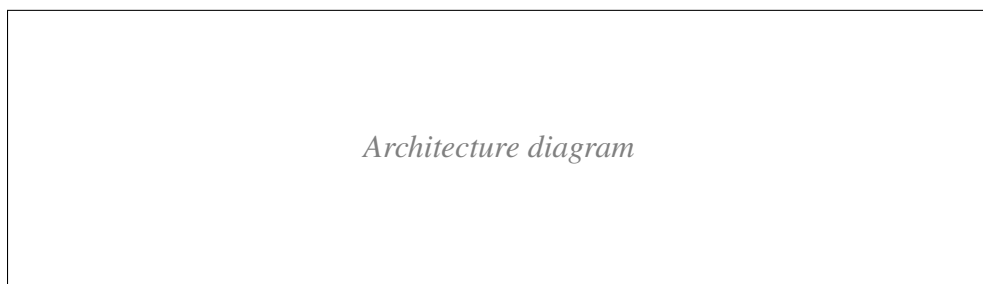


Figure 4.1: System architecture

Partition the problem into sub-problems or modules; explain each one's responsibility and how they interact. Show where the standards and codes of practice from Section 3.2 and any health, safety, or environmental considerations shaped the design.

4.2 Design Refinement and Simulation

Present the detailed design using the views that fit the project (ERD, class, sequence, DFD, API, wireframes, circuit, model). Include any design calculations or estimates made (e.g. expected load or storage), the analyses, prototypes, or simulations used to verify the design against the requirements, and how the design was refined to facilitate implementation.

4.3 Application of Tools

Justify the selection of the important tools and technologies and state their relevant limitations.

4.4 Implementation

Describe how the key modules were implemented and the development process followed (repository, version control, code review, CI/CD).

4.5 Deployment

Describe how the system is deployed and operated: the target environment and infrastructure, configuration and release process, monitoring and backup, and who operates and maintains it. State the computing resources the deployment consumes; their environmental impact is assessed in Section 6.4.

Chapter 5

Investigation and Evaluation

5.1 Experimental Investigation

Present the investigation carried out to substantiate the solution. State the research questions or hypotheses, the method and experimental design, the datasets or measurements used and how they were obtained, the baselines or points of comparison, and the metrics. Analyse and interpret the data, and synthesise the findings into conclusions that are supported by the evidence. Projects with a substantial research and development component should treat this as the core of the chapter; for development-focused projects, report the investigation that validated the main design decisions. This is the experimental work on the built system; the background research is in Chapter 2.

5.2 Performance Evaluation

Evaluate the system against every requirement and objective. Describe the setup, tools, inputs, and acceptance criteria; present the results (tests, measurements, security checks, user or partner acceptance, experiments as appropriate); state which requirements are met and clearly explain each deviation.

5.3 Deviations and Design Revisions

For each deviation, give the technical analysis of its cause and the design revision made to achieve compliance, or explain why it was not revised.

Chapter 6

Project Management, Teamwork, and Impact

6.1 Project Management and Finance

6.1.1 Planning and Risk Management

Present the plan: phases, milestones, task distribution, and schedule, noting where actual progress differed. List the major risks and how they were mitigated.

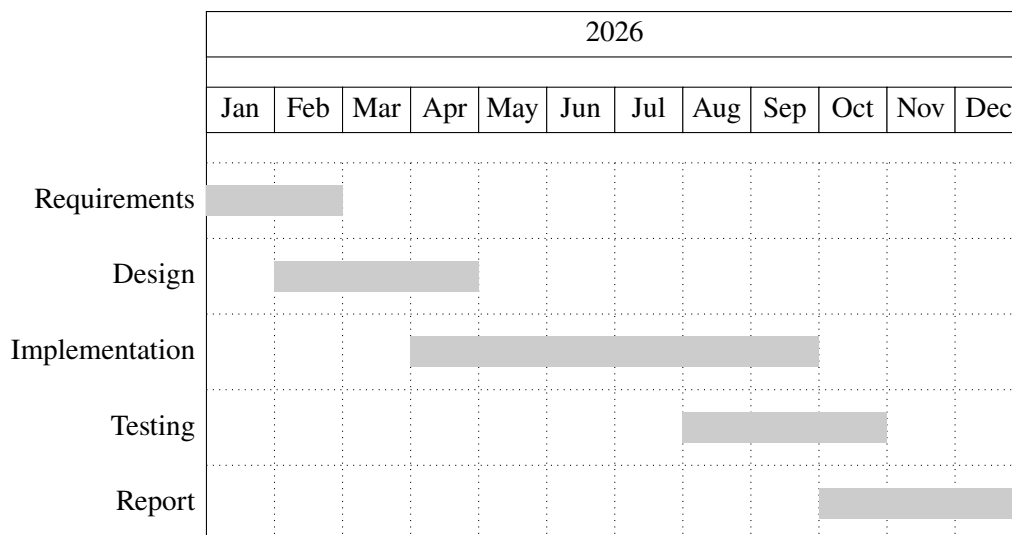


Figure 6.1: Project schedule

6.1.2 Budgeting and Resource Identification

Identify the resources required (people, hardware, software, cloud and API services, data) with an itemised budget.

6.1.3 Economic Analysis and Financial Prospecting

Evaluate the financial prospect of the project with clear interpretation: development and operating cost, the potential users or market established in Section 2.3, and how the solution could generate value or revenue.

6.1.4 Sustainability in Business and Commercialisation

Explain how the solution could remain viable after the course: the commercialisation or deployment path (e.g. with the industry partner) and who would maintain and operate it.

6.2 Individual and Teamwork

6.2.1 Participation and Contribution

For each member, give the role, specific contributions, participation in discussions and decision-making, and whether responsibilities were completed on time.

6.2.2 Collaboration and Conflict Resolution

Describe how the team collaborated and kept a constructive atmosphere, and how any conflicts were identified and resolved.

6.2.3 Leadership and Direction

Explain how direction toward the goals was provided, how members' views were listened to, how the team planned for improvement after setbacks, and how members were motivated.

6.2.4 Multidisciplinary Engagement

Explain the multidisciplinary nature of the project and the team's participation in activities outside its own discipline: work with the industry partner, domain experts, end users, or other departments.

6.3 Impact on Society

Consider all impacts of the solution on society, positive and negative, covering health, safety, cultural, and legal issues explicitly. For each, state who is affected and how the design responds. Be specific to this project.

6.4 Environmental Impact and Life Cycle Sustainability

Consider the environmental and sustainability impacts over the entire life cycle: development, deployment and operation, and retirement of the system and its data. Identify the negative impacts and propose mitigation for each.

6.5 Ethics

6.5.1 Equity and Inclusivity

Show how equity and inclusivity were considered: accessibility, fairness across user groups, and who might be excluded and how that was addressed.

6.5.2 Accountability and Personal Responsibility

Show accountability and personal responsibility: who is responsible for the system's behaviour and failures, and how members took responsibility for their own work.

6.5.3 Proper Use of Intellectual Property

Show due acknowledgement of others' work, prior art, intellectual property, and copyrighted material: cite sources, list third-party code, models, and datasets with their licences, and disclose any generative AI use.

6.5.4 Professionalism and Ethical Codes

Show understanding of the engineer's role in protecting the public interest and awareness of the relevant codes of ethics (e.g. ACM, IEEE, IEB). Explain how they guided specific project decisions.

Chapter 7

Conclusion

7.1 Conclusion and Future Work

Summarise achievement against each objective, the limitations, and focused future work. Do not repeat the abstract.

7.2 Continuous Engagement and Self-directed Learning

Demonstrate engagement in staying up to date: the tools, technologies, and domain knowledge learned independently during the project, and the emerging technical literature or documentation identified and used.